

Edge Detection using Sobel Filter

$$\begin{bmatrix} 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \end{bmatrix}_{5 \times 5}$$

$$\begin{aligned} \text{Output size} &= \frac{N - F + 2P}{S} + 1 \\ &= \frac{5 - 3 + 0}{1} + 1 \\ &= \underline{\underline{3}} \end{aligned}$$

where

$N \Rightarrow$ Input image size = 5

$F \Rightarrow$ Filter size = 3

$P \Rightarrow$ Padding = 0

$S = \text{Stride} = 1$

$$\text{Output size} = \underline{\underline{3 \times 3}}$$

$$\text{Sobel } X = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

1st Convolution:

$$\text{Image Patch} = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}; \text{Sobel } X = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{Output}(1,1) &= (-10 + 0 + 10) + (-20 + 0 + 20) + (-10 + 0 + 10) \\ &= \underline{\underline{0}} \end{aligned}$$

Output (1,2) =

$$\begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{Output (1,2)} &= (-1 \times 10) + (0) + (80) + \\ &\quad (-20 + 0 + 160) + \\ &\quad (-10 + 0 + 80) = 280 \end{aligned}$$

$$\text{Output (1,2)} = 280.$$

Output (1,3);

$$\begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\text{Output (1,3)} = (-10 + 0 + 80) + (-20 + 0 + 160) + (-10 + 0 + 80)$$

$$\text{Output (1,3)} = 280$$

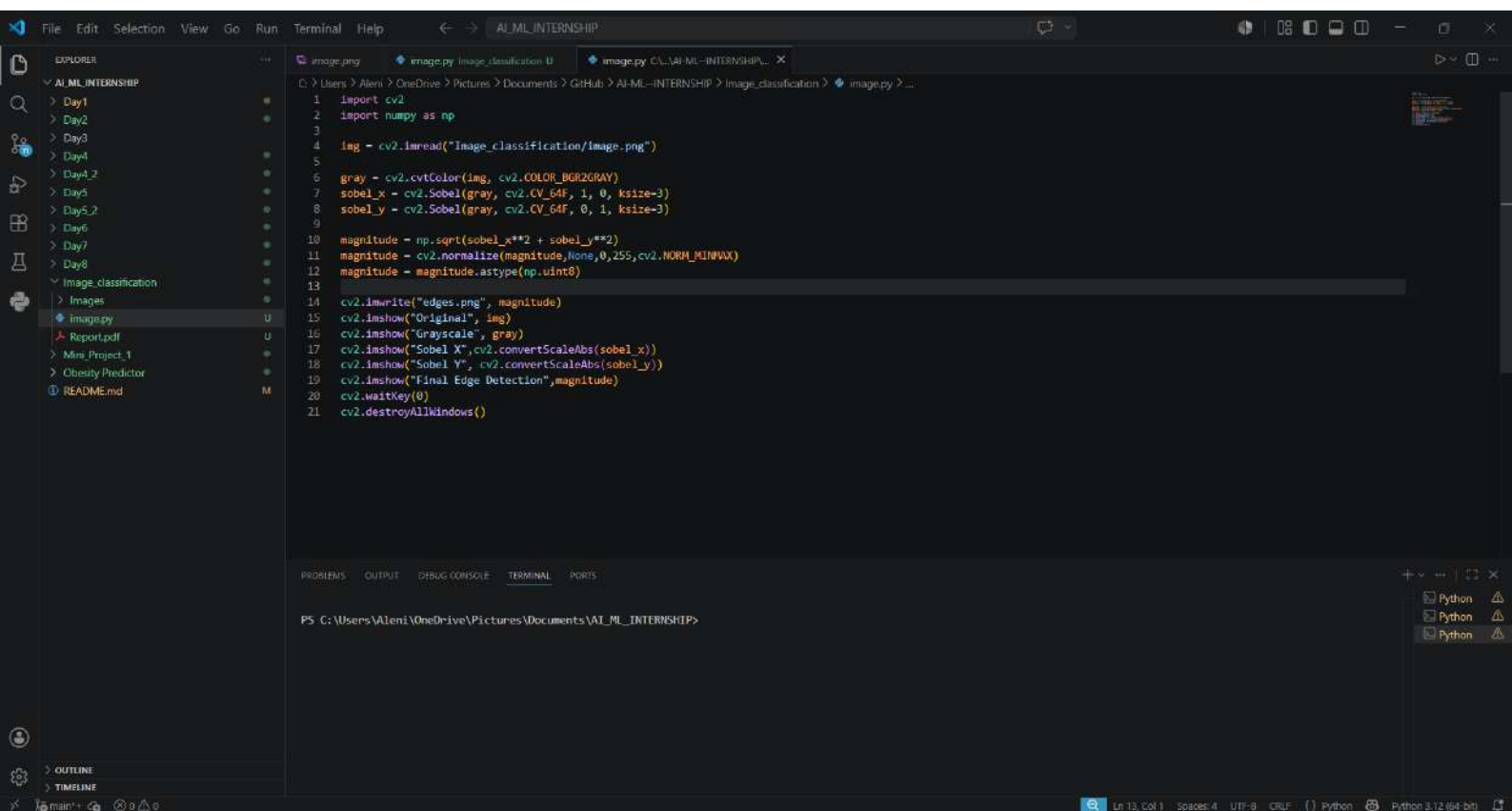
As all the rows are identical;

The remaining rows will give same answer

$$\text{Final Output} = \begin{bmatrix} 0 & 280 & 280 \\ 0 & 280 & 280 \\ 0 & 280 & 280 \end{bmatrix}$$

Analysis:

The Sobel-X operator detects vertical edges. The given image intensity changes from 10 to 80 b/w 3rd and 4th columns, producing a strong response of 280. Homogeneous regions produce value close to zero because positive and negative weights cancel each other.



Original Image



Grayscale



Sobel X



Sobel Y



Edge Detection

